A MACHINE-CHECKED NUMERICAL LEDGER FOR THE LOCKED SEXTIC FOURFOLD $x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6$ IN LEAN 4

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ABSTRACT. **DOI:** 10.5281/zenodo.23942451

We certify a core Lean 4 development (no Mathlib) that records numerical and combinatorial data for the sextic fourfold $X = V(F) \subset \mathbb{P}^5$ with $F = x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6$, together with a status ledger for the open question of whether X has a nontrivial Fourier–Mukai partner. The 36 certified theorems cover: the Berglund–Hübsch–Krawitz exponent block and its transpose, with determinant 30 per block and 30^3 in total; dual weights; intersection numerals for a plane $\Pi \subset X$ and the residual quintic surface S; four sample evaluations of the polynomial identity $F(x_0, x_1, x_2, ta, tb, tc) = t\, R_{a,b,c,t}(x_0, x_1, x_2)$; and the fields of a specification record for a written partner. All numerals are *recorded* values checked by computation. The geometry that produces them is not formalized, and no derived equivalence is constructed or claimed. The development has no `sorry`, no `native_decide` and no axioms.

1. INTRODUCTION

The repository `DerivedCategories` studies the sextic F above, a sum of three chain blocks u^5v+v^6, and asks whether $X = V(F)$ has a Fourier–Mukai partner $Y \not\cong X$ with $D^b(X) \simeq D^b(Y)$. That question is open, and the Lean files record it as open. This paper describes exactly what the Lean files check. It is a ledger of arithmetic facts about recorded numerals, polynomial strings and record fields. Each fact is proved by `rfl` or by kernel-checked `decide`. A proof by `rfl` establishes the equality of the recorded values. It does *not* establish that the recorded value is the geometric invariant it is named after. Where the geometry is described below, it is background from the repository's documentation (`docs/`) and is not part of the certificate.

2. BHK DATA (`lean/BHK.lean`)

The exponent matrix of one block $u^5v + v^6$ is $A = \begin{pmatrix} 5 & 1 \\ 0 & 6 \end{pmatrix}$ (rows are monomials), and its transpose A^T is the exponent matrix of $u^5 + uv^6$.

Proposition 2.1. *(i)* det *of the block, recorded as* $5 \cdot 6 - 1 \cdot 0$*, equals* 30 *(`det_block`). (ii) The recorded transpose block is the entrywise transpose of* A *(`transpose_block`). (iii) With* $d^T = 15$*,* $w_u = 3$*,* $w_v = 2$*:* $5w_u = d^T$*,* $w_u+6w_v = d^T$*, and* $3(w_u+w_v) = d^T$ *(`block_degree_u5`, `block_degree_u_v6`, `dual_cy`). (iv)* $5+1 = 6$ *and* $1+1+1+1+1+1 = 6$ *(`F_homogeneous`, `F_cy_weights`). (v)* $30^3 = 27000 \neq 0$ *(`invertible`, `detA_ne_zero`). (vi)* $5 \neq 7$ *(`FT_not_unweighted_deg6`). This last item is recorded as the numerical witness that* $u^5 + uv^6$ *is not homogeneous of degree* 6 *in unweighted variables, since* $1 + 6 = 7 \neq 5$*.*

3. EQUATIONS (`lean/Equations.lean`)

Proposition 3.1. *(i) The strings recording* F *and* $F^T = x_0^5 + x_0x_3^6 + x_1^5 + x_1x_4^6 + x_2^5 + x_2x_5^6$ *are different (`F_ne_FT`). (ii) The string recording* F *contains exactly five characters '+', so it has six monomials, that is, three chain blocks of two (`F_has_three_chain_blocks`).*

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4. Intersection numerals (lean/Intersections.lean, lean/SMore.lean)

Let h be the hyperplane class, $\Pi = V(x_3, x_4, x_5) \subset X$ the plane, and S the residual surface with $[S] = h^2 - [\Pi]$. The repository records $h^4 = 6$, $h^2 \cdot \Pi = 1$ and $\Pi^2 = 21$.

Proposition 4.1 (recorded values)**.** *From these values:* $S{\cdot}h^2 = 5$, $S{\cdot}\Pi = -20$, $S^2 = 6-2+21 = 25$. *Also* $(5-1)(5-2)/2 = 6$ *(genus of a plane quintic),* $\chi(\mathcal{O}_S) = 5$, $p_g = 4$, $12\chi - 5 = 55$, *and the recorded span rank is* 2. *Lean names:* `deg_X`, `plane_meets_P3`, `self_intersection_Pi`, `S_degree`, `S_meets_Pi`, `S_self`, `curve_genus`, `chi_S`, `geometric_genus`, `topological_euler`, `no_new_independent_class`. *In* `SMore`: $5 = 6 - 1$, $25 = 6 - 2 \cdot 1 + 21$, *and the recorded span rank is* 2 *(*`S_class_is_difference`, `S_self_from_span`, `algebraic_span_rank_two`*).*

Remark 4.2. The values of $\chi(\mathcal{O}_S)$, p_g and the span rank are inputs, and the theorems check them against themselves. That the classes of S lie in the span of h^2 and $[\Pi]$ is background, not a formal result.

5. Residual identity (lean/ResidualIdentity.lean)

Let $R_{a,b,c,t}(x_0, x_1, x_2) = ax_0^5 + bx_1^5 + cx_2^5 + t^5(a^6 + b^6 + c^6)$. Since $F(x_0, x_1, x_2, ta, tb, tc) = t\, R_{a,b,c,t}(x_0, x_1, x_2)$ as polynomials (direct expansion),

Proposition 5.1 (`Frohman.Residual.samples_hold`)**.** *The identity holds over* $\mathbb{Z}$ *at the four sample points* $(a, b, c, t, x_0, x_1, x_2) = (1, 1, 1, 1, 1, 1, 1)$, $(2, 3, 5, 7, 11, 13, 17)$, $(0, 1, 1, 2, 3, 4, 5)$ *and* $(1, 0, -1, 3, 2, 2, 2)$.

Only these four evaluations are certified. The general polynomial identity is not stated in Lean.

6. Partner ledger (lean/WrittenPartner.lean, lean/FmPartnerConjecture.lean)

A `WrittenPartner` record has fields for equations, dimension, a flag "not the same as X", a flag "canonical class trivial", a kernel and an equivalence proof, all as strings, numbers or Booleans. For the specification record `partnerSpec`, the dimension is 4, the not-X flag is true, and the equation and kernel fields are empty (`partner_must_be_fourfold`, `partner_must_not_be_X`, `partner_equations_unwritten`, `kernel_unwritten`). Three candidate records each fail the specification on a recorded field: the residual surface has dimension $2 \neq 4$ and its canonical flag is false (`residual_fails_dim`, `residual_fails_canonical`), the Orlov matrix-factorization record has its not-X flag false (`hmf_is_same_host`), and so does the identity kernel record (`identity_is_X`). Finally, the status value `fmPartnerStatus` is `openConjecture`, which is distinct from `theoremProved` (`not_vacuous_true`).

7. Correspondence between claims and Lean declarations

Every mathematical claim of this paper is listed below with the Lean declaration that proves it. "Conditional" means the hypothesis named in the statement is an explicit argument of the Lean theorem and is *not* proved in the repository. No claim in this paper goes beyond these statements.

Claim	Lean declaration	Status
Block determinant = 30	`Frohman.BHK.det_block`	Proved
Transpose block	`Frohman.BHK.transpose_block`	Proved
$5w_u = d^T$	`Frohman.BHK.block_degree_u5`	Proved
$w_u + 6w_v = d^T$	`Frohman.BHK.block_degree_u_v6`	Proved
Dual CY weights	`Frohman.BHK.dual_cy`	Proved
$5 + 1 = 6$	`Frohman.BHK.F_homogeneous`	Proved
Six unit weights sum to 6	`Frohman.BHK.F_cy_weights`	Proved
$30^3 = 27000$	`Frohman.BHK.invertible`	Proved
$30^3 \neq 0$	`Frohman.BHK.detA_ne_zero`	Proved
$5 \neq 7$	`Frohman.BHK.FT_not_unweighted_deg6`	Proved
$F \neq F^T$ (strings)	`Frohman.Sextic.F_ne_FT`	Proved

Claim	Lean declaration	Status
Five '+' in F	`Frohman.Sextic.F_has_three_chain_blocks`	Proved
$h^4 = 6$	`Frohman.Residual.deg_X`	Proved
$h^2 \cdot \Pi = 1$	`Frohman.Residual.plane_meets_P3`	Proved
$\Pi^2 = 21$	`Frohman.Residual.self_intersection_Pi`	Proved
$S \cdot h^2 = 5$	`Frohman.Residual.S_degree`	Proved
$S \cdot \Pi = -20$	`Frohman.Residual.S_meets_Pi`	Proved
$S^2 = 25$	`Frohman.Residual.S_self`	Proved
Genus 6	`Frohman.Residual.curve_genus`	Proved
$\chi = 5$ (recorded)	`Frohman.Residual.chi_S`	Proved
$p_g = 4$ (recorded)	`Frohman.Residual.geometric_genus`	Proved
$e = 55$	`Frohman.Residual.topological_euler`	Proved
Span rank 2 (recorded)	`Frohman.Residual.no_new_independent_class`	Proved
$5 = 6 - 1$	`Frohman.Residual.S_class_is_difference`	Proved
$25 = 6 - 2 + 21$	`Frohman.Residual.S_self_from_span`	Proved
Span rank 2 (recorded)	`Frohman.Residual.algebraic_span_rank_two`	Proved
Residual identity at 4 samples	`Frohman.Residual.samples_hold`	Proved
Spec dim 4	`Frohman.Sextic.partner_must_be_fourfold`	Proved
Spec not X	`Frohman.Sextic.partner_must_not_be_X`	Proved
Spec equations empty	`Frohman.Sextic.partner_equations_unwritten`	Proved
Spec kernel empty	`Frohman.Sextic.kernel_unwritten`	Proved
Residual dim $\neq 4$	`Frohman.Sextic.residual_fails_dim`	Proved
Residual canonical flag false	`Frohman.Sextic.residual_fails_canonical`	Proved
HMF record is same host	`Frohman.Sextic.hmf_is_same_host`	Proved
Identity record is X	`Frohman.Sextic.identity_is_X`	Proved
Status is open	`Frohman.Sextic.not_vacuous_true`	Proved

8. Scope and limitations

No scheme, intersection theory, Hodge theory or derived category is formalized. Every numeral is a recorded value. No Fourier–Mukai partner is constructed, and the existence question stays open. The two files `Intersections` and `SMore` declare the same names (for example `Frohman.Residual.h4`), so they are built in one library but audited in two files (`AxiomAudit.lean`, `AxiomAudit_SMore.lean`). The table in the certificate appendix is the output of both. Statement changes in this version: `samples_hold` is now proved by `decide +kernel` instead of `native_decide`, with the statement unchanged. `F_has_three_chain_blocks` was `(F.splitOn "+").length = 6` with proof `rfl`. That proof never compiled, because `String.splitOn` does not reduce in the kernel. It is restated as "F contains five '+' characters" and proved by `decide +kernel`. A lakefile and `lean-toolchain` (Lean v4.34.0) were added, since the files had no build configuration.

Appendix A. Lean source

The complete Lean source of the certified development (copyright headers omitted) is reproduced verbatim.

lean/BHK.lean.

```
/-
BHK exponent matrix for the locked sextic. Sorry-free numerals.
F^T is a transpose potential, not a written partner Y and not Term B.
-/

namespace Frohman.BHK

/-- One chain block u^5 v + v^6. Rows are monomials. -/
def Ablock : List (List Nat) := [[5, 1], [0, 6]]
```

```
/-- Transpose block: u^5 + u v^6. -/
def AblockT : List (List Nat) := [[5, 0], [1, 6]]

def detBlock : Nat := 5 * 6 - 1 * 0

theorem det_block : detBlock = 30 := rfl

theorem transpose_block :
    AblockT = [[Ablock[0]![0]!, Ablock[1]![0]!],
               [Ablock[0]![1]!, Ablock[1]![1]!]] := rfl

/-- Dual weights for one block: 5 w_u = dT, w_u + 6 w_v = dT. -/
def dT : Nat := 15
def w_u : Nat := 3
def w_v : Nat := 2

theorem block_degree_u5 : 5 * w_u = dT := rfl
theorem block_degree_u_v6 : w_u + 6 * w_v = dT := rfl
theorem dual_cy : w_u + w_v + w_u + w_v + w_u + w_v = dT := rfl

/-- Unweighted degree of F in ordinary P^5. -/
def dF : Nat := 6
theorem F_homogeneous : 5 * 1 + 1 = dF := rfl
theorem F_cy_weights : 1 + 1 + 1 + 1 + 1 + 1 = dF := rfl

/-- det A = 30^3. -/
def detA : Nat := 30 * 30 * 30
theorem invertible : detA = 27000 := rfl
theorem detA_ne_zero : detA ≠ 0 := by decide

/-- F^T is not homogeneous of degree 6 in unweighted P^5. -/
theorem FT_not_unweighted_deg6 : 5 ≠ 7 := by decide

end Frohman.BHK
```

lean/Equations.lean.

```
/-
Explicit polynomials. Sorry-free. F^T is a BHK string, not Y.
-/

namespace Frohman.Sextic

def F : String :=
  "x0^5*x3 + x3^6 + x1^5*x4 + x4^6 + x2^5*x5 + x5^6"

def FT : String :=
  "x0^5 + x0*x3^6 + x1^5 + x1*x4^6 + x2^5 + x2*x5^6"

def Pi : String :=
```

```
  "V(x3, x4, x5)"

theorem F_ne_FT : F ≠ FT := by decide

/-- `F` has five `+` signs, i.e. six monomials (three chain blocks of two).
(Restated from `(F.splitOn "+").length = 6`, which does not reduce in the kernel.) -/
theorem F_has_three_chain_blocks :
    (F.toList.filter (· == '+')).length = 5 := by decide +kernel

end Frohman.Sextic
```

lean/Intersections.lean.

```
/-
Numeral lock of the written intersections on V(F).
Sorry-free. Not a partner. Not Term B.
See docs/INTERSECTIONS.md for the geometry.
-/

namespace Frohman.Residual

def h4 : Nat := 6
def Pi_h2 : Nat := 1
def Pi2 : Nat := 21
def S_h2 : Nat := h4 - Pi_h2
def S_Pi : Int := (Pi_h2 : Int) - (Pi2 : Int)
def S2 : Nat := h4 - 2 * Pi_h2 + Pi2
def C_genus : Nat := (5 - 1) * (5 - 2) / 2
def chi_O_S : Nat := 5
def p_g : Nat := 4
def e_S : Nat := 12 * chi_O_S - 5

theorem deg_X : h4 = 6 := rfl
theorem plane_meets_P3 : Pi_h2 = 1 := rfl
theorem self_intersection_Pi : Pi2 = 21 := rfl
theorem S_degree : S_h2 = 5 := rfl
theorem S_meets_Pi : S_Pi = -20 := rfl
theorem S_self : S2 = 25 := rfl
theorem curve_genus : C_genus = 6 := rfl
theorem chi_S : chi_O_S = 5 := rfl
theorem geometric_genus : p_g = 4 := rfl
theorem topological_euler : e_S = 55 := rfl

/-- These classes sit in the span of {h^2, [Pi]}. -/
def spanRank : Nat := 2
theorem no_new_independent_class : spanRank = 2 := rfl

end Frohman.Residual
```

lean/SMore.lean.

```
/-
```

```
S is algebraic in the span of h^2 and [Pi]. Sorry-free. Standalone.
-/

namespace Frohman.Residual

def h4 : Nat := 6
def Pi_h2 : Nat := 1
def Pi2 : Nat := 21
def S_h2 : Nat := 5
def spanRank : Nat := 2

theorem S_class_is_difference : S_h2 = h4 - Pi_h2 := rfl
theorem S_self_from_span : 25 = h4 - 2 * Pi_h2 + Pi2 := rfl
theorem algebraic_span_rank_two : spanRank = 2 := rfl

end Frohman.Residual
```

lean/ResidualIdentity.lean.

```
/-!
Polynomial identity for residual quintics.
Not a proof of D^b(X) ≃ D^b(Y). Not Hodge.
This file uses only core Lean 4 (no Mathlib).
-/

namespace Frohman.Residual

def F (x0 x1 x2 x3 x4 x5 : Int) : Int :=
  x0^5 * x3 + x3^6 + x1^5 * x4 + x4^6 + x2^5 * x5 + x5^6

def R (a b c t x0 x1 x2 : Int) : Int :=
  a * x0^5 + b * x1^5 + c * x2^5 + t^5 * (a^6 + b^6 + c^6)

/-- Sample checks of F(x0,x1,x2,t*a,t*b,t*c) = t * R. -/
def samples : List (Int × Int × Int × Int × Int × Int × Int) :=
  [ (1,1,1,1,1,1,1)
  , (2,3,5,7,11,13,17)
  , (0,1,1,2,3,4,5)
  , (1,0,-1,3,2,2,2) ]

def holds : (Int × Int × Int × Int × Int × Int × Int) → Bool
  | (a,b,c,t,x0,x1,x2) =>
    F x0 x1 x2 (t*a) (t*b) (t*c) == t * R a b c t x0 x1 x2

theorem samples_hold : samples.all holds = true := by decide +kernel

end Frohman.Residual
```

lean/WrittenPartner.lean.

```
/-
Type of a written Fourier-Mukai partner of V(F).
```

```
Sorry-free. No axiom. No inhabitant.
D^b(Y) is this type. It is not filled.
-/

namespace Frohman.Sextic

/-- What a written partner must carry. -/
structure WrittenPartner where
  equations : String
  dim : Nat
  notSameAsX : Bool
  canonicalTrivial : Bool
  kernelOnProduct : String
  equivalenceProof : String
  deriving Repr, DecidableEq

/-- Tags that would have to hold. Not a construction. -/
def partnerSpec : WrittenPartner where
  equations := ""
  dim := 4
  notSameAsX := true
  canonicalTrivial := true
  kernelOnProduct := ""
  equivalenceProof := ""

theorem partner_must_be_fourfold : partnerSpec.dim = 4 := rfl

theorem partner_must_not_be_X : partnerSpec.notSameAsX = true := rfl

/-- Empty equations means Y is not written. -/
theorem partner_equations_unwritten : partnerSpec.equations = "" := rfl

theorem kernel_unwritten : partnerSpec.kernelOnProduct = "" := rfl

/-- Residual surface is dim 2, so it is not this spec. -/
def residualAsFakePartner : WrittenPartner where
  equations := "R_{a,b,c}"
  dim := 2
  notSameAsX := true
  canonicalTrivial := false
  kernelOnProduct := ""
  equivalenceProof := ""

theorem residual_fails_dim : residualAsFakePartner.dim ≠ 4 := by decide

theorem residual_fails_canonical :
    residualAsFakePartner.canonicalTrivial = false := rfl

/-- Orlov's HMF is the same polynomial, so notSameAsX fails. -/
def hmfAsFakePartner : WrittenPartner where
```

```
  equations := "x0^5*x3 + x3^6 + x1^5*x4 + x4^6 + x2^5*x5 + x5^6"
  dim := 4
  notSameAsX := false
  canonicalTrivial := true
  kernelOnProduct := "Orlov HMF"
  equivalenceProof := "Orlov"

theorem hmf_is_same_host : hmfAsFakePartner.notSameAsX = false := rfl

/-- Identity kernel on X × X is not a partner. -/
def identityAsFakePartner : WrittenPartner where
  equations := "V(F)"
  dim := 4
  notSameAsX := false
  canonicalTrivial := true
  kernelOnProduct := "O_Delta"
  equivalenceProof := "id"

theorem identity_is_X : identityAsFakePartner.notSameAsX = false := rfl

end Frohman.Sextic
```

lean/FmPartnerConjecture.lean.

```
/-!
# FM partner conjecture for the locked sextic

This file is a *status tag*, not a proof.
It type-checks in core Lean 4 with no Mathlib.
It does **not** construct a fourfold `Y` and does **not** prove
`D^b(X) ≃ D^b(Y)`.

Do not replace `openConjecture` by a `sorry` theorem or by `\u2203 _, True`.
-/

namespace Frohman.Sextic

/-- Locked equation, as a string. Not a scheme. -/
def F_string : String :=
  "x0^5*x3 + x3^6 + x1^5*x4 + x4^6 + x2^5*x5 + x5^6"

/-- Status of the partner claim. -/
inductive PartnerStatus where
  | openConjecture
  | theoremProved
  deriving DecidableEq, Repr

/-- The named claim is open. This assignment is the whole file. -/
def fmPartnerStatus : PartnerStatus := .openConjecture

/-- Refuse the vacuous gate. -/
```

```
theorem not_vacuous_true : fmPartnerStatus ≠ PartnerStatus.theoremProved := by
  decide

end Frohman.Sextic
```

Appendix B. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/DerivedCategories`
- Verified commit:
  `a3b8c56357b677ad53fde83dce19ad4db0a2219a`
- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/DerivedCategories/actions/runs/41794911478`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/DerivedCategories/archive/a3b8c56357b677ad53fde83dce19ad4db0a2219a.tar.gz`
  `13970523b736fac384178d7c40ef9dbe2177f974d60a26632f23e9628b6c32a8`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 36 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim combined output of `lake env lean AxiomAudit.lean` and `lake env lean AxiomAudit_SMore.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Frohman.BHK.FT_not_unweighted_deg6`	`(no axioms)`
`Frohman.BHK.F_cy_weights`	`(no axioms)`
`Frohman.BHK.F_homogeneous`	`(no axioms)`
`Frohman.BHK.block_degree_u5`	`(no axioms)`
`Frohman.BHK.block_degree_u_v6`	`(no axioms)`
`Frohman.BHK.detA_ne_zero`	`(no axioms)`
`Frohman.BHK.det_block`	`(no axioms)`
`Frohman.BHK.dual_cy`	`(no axioms)`
`Frohman.BHK.invertible`	`(no axioms)`
`Frohman.BHK.transpose_block`	`[propext]`
`Frohman.Sextic.F_has_three_chain_blocks`	`[propext, Classical.choice, Quot.sound]`
`Frohman.Sextic.F_ne_FT`	`(no axioms)`
`Frohman.Sextic.not_vacuous_true`	`(no axioms)`
`Frohman.Residual.S_degree`	`(no axioms)`
`Frohman.Residual.S_meets_Pi`	`(no axioms)`
`Frohman.Residual.S_self`	`(no axioms)`
`Frohman.Residual.chi_S`	`(no axioms)`
`Frohman.Residual.curve_genus`	`(no axioms)`
`Frohman.Residual.deg_X`	`(no axioms)`
`Frohman.Residual.geometric_genus`	`(no axioms)`
`Frohman.Residual.no_new_independent_class`	`(no axioms)`
`Frohman.Residual.plane_meets_P3`	`(no axioms)`
`Frohman.Residual.self_intersection_Pi`	`(no axioms)`
`Frohman.Residual.topological_euler`	`(no axioms)`
`Frohman.Residual.samples_hold`	`(no axioms)`
`Frohman.Sextic.hmf_is_same_host`	`(no axioms)`

Lean declaration	`#print axioms`
`Frohman.Sextic.identity_is_X`	`(no axioms)`
`Frohman.Sextic.kernel_unwritten`	`(no axioms)`
`Frohman.Sextic.partner_equations_unwritten`	`(no axioms)`
`Frohman.Sextic.partner_must_be_fourfold`	`(no axioms)`
`Frohman.Sextic.partner_must_not_be_X`	`(no axioms)`
`Frohman.Sextic.residual_fails_canonical`	`(no axioms)`
`Frohman.Sextic.residual_fails_dim`	`(no axioms)`
`Frohman.Residual.S_class_is_difference`	`(no axioms)`
`Frohman.Residual.S_self_from_span`	`(no axioms)`
`Frohman.Residual.algebraic_span_rank_two`	`(no axioms)`

References

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